



ANDY WANG
SENIOR BUSINESS MANAGER,
ENERGY SOLUTIONS BUSINESS
GROUP, MAXIM INTEGRATED

The deployment of smart meters is far from a one-size-fits-all undertaking

Highly integrated SoCs are helping design energy meters with ease, but what are the design considerations and challenges for energy metering? And how can you select an SoC for designing energy meters? Andy Wang, senior business manager, Energy Solutions Business Group, Maxim Integrated talks to Pankaj Vashisht from EFY

Q. What are the key elements of smart meter design?

The ability of solid-state electricity meters to detect and prevent tampering can significantly improve control and cost recovery for utility companies. By enabling customers to better manage their own energy usage through incentive-based programs—such as direct load control, interruptibility rate agreements and demand bidding or buyback—smart metering can help utilities manage overall energy consumption patterns and cope with peak-demand challenges.

Q. What are the biggest considerations and challenges in designing these meters?

The deployment of smart meters is far from a one-size-fits-all undertaking. Manufacturers must account for the varying regulatory requirements of each region as well as the different functionalities and services required for different markets. For servicing global markets, the combination of different driving forces or elements presents significant opportunities and challenges. Therefore meter manufacturers need to be flexible, offering both low-cost metering solutions and high-end smart meter alternatives.

The biggest challenge in designing any energy meter is to make sure the metrology is stable and repeatable in terms of accuracy. When it comes to designing energy meters, the accuracy of the integrated smart devices is also very important, because it decides the reliability of the system.

Q. What are the benefits provided by highly integrated SoCs in designing energy meters?

Using these SoCs, one doesn't need to spend much on engineering resources. This is because the engineer does not have to start the design from scratch. SoCs lower the bill of material costs and provide rich feature sets for smart metering, and simpler upgrade paths with minimum hardware and operational costs. Their key strength is the family effect and scalability. Engineers can design with one product and easily migrate to different products in the family depending on their needs.

Q. What ensures stability and repeatability in terms of accuracy in energy meters?

The basic features like the dynamic range of the ADC, sample rate of the ADC and the processing power of the

microcontrollers account for the accuracy in energy metering. Also, the metrology compute engine (CE) ensures high-accuracy processing of all collected data. It is dedicated to computing the metering parameters from voltage and current samples.

Q. What new features do these SoCs bring to energy metering?

If you look at the whole family from low-end market to high-end market, when you go to the higher end, you will get more memory, which in turn allows the engineer to incorporate DLMS/COSEM stacks (communication protocols for smart meter), allowing you to communicate with the modems. Also, in some regions it is required to encrypt/decrypt the data, and for that these SoCs incorporate hardware accelerators.

Q. Any key parameters for selecting a SoC for energy-metering requirement?

Accuracy, memory size, processing power and amount of peripherals integrated are the key parameters to consider when selecting any metering SoC solution. An engineer should make sure it features field programmability, enables ease of integration with local devices such as thermostats, and supports a variety of sensor inputs with minimum hardware.

Q. What are the tools and guidelines available to design with SoCs?

From hardware perspective, there are reference designs, layout and hardware guidelines. All one has to do is leverage the technology with these SoCs and modify it, and just make sure the power supplied is stable and sensing circuitry is clean. From software perspective, we provide modulated source code, so one can take it from there and easily modify it depending upon end customer's requirements.

Q. How do you see the future of energy metering?

Looking into the future, I feel it is about more upgrades in the metrology along with the metering firmware. Products that most vendors vision are the products with increased integration of communications and security features. ●